Continuous Convergence and Curried Functional Spaces

1. The language of filters

Definition 1.1. Let X be a set. A *filter* on X is a non-empty set $F \subset 2^X$ which is closed under supersets and finite intersections, and does not contain the empty set. The set of all filters on X will be denoted by $\mathcal{F}(X)$.

Definition 1.2. Let X be a set. A family $\mathcal{B} \subset 2^X$ is called a *filter base* if it does not contain the empty set and is closed under finite intersection. The filter $[\mathcal{B}]$ *generated* by $\mathcal{B}$ is defined as the collection of all supersets of sets from $\mathcal{B}$.
If $\mathcal{B} = \{\{x\}\}$ where $x \in X$, the filter $[x] = [\{\{x\}\}]$ is called the *universal filter* of x.
If $f : X \to Y$ is a map and $F \in \mathcal{F}(X)$ is a filter on X, then $\{f(A) \mid A \in F\}$ is a filter base that generates the *image filter* denoted $f(F)$.

Definition 1.3. Let X be a topological space. Then we say that a filter F on X *converges* to a point $x \in X$, and write $F \to x$, if it contains all open neighborhoods of x.

Remark 1.1. The language of filters is expressive enough to encode all topological properties. For example [1]:

- A sequence $\{x_n\}_{n=1}^{\infty}$ converges to a point $x \in X$ iff the filter
$$F = \{A \subset X \mid x_n \in A \text{ for all but finitely many } n\}$$
converges to x. This filter F is called the *derived filter* of $\{x_n\}_{n=1}^{\infty}$.
- A subset $E \subset X$ is closed if for every point $x \in E$ there is a filter $F \in \mathcal{F}(X)$ converging to x such that $A \cap E \neq \varnothing$ for all $A \in F$.
- A topological space X is compact iff every filter F on X is contained in a convergent filter on X.
- A function $f : X \to Y$ between topological spaces is continuous iff it preserves convergent filters, i.e. $F \to x \in X$ implies $f(F) \to f(x) \in Y$.

2. Curried functions

Definition 2.1. Let $n \in \mathbb{N}_0 = \{0, 1, 2, ...\}$. The set $H_n(\mathbb{R})$ of *curried n-variable functions* is a topological vector space defined as follows:

- If $n = 0$, we set $H_0(\mathbb{R}) = \mathbb{R}$;
- If $H_n(\mathbb{R})$ is defined, we define $H_{n+1}(\mathbb{R})$ to be the set of all functions from $\mathbb{R}$ to $H_n(\mathbb{R})$. This set is given the pointwise topological and linear structure. In other words, a filter $F \in \mathcal{F}(H_{n+1}(\mathbb{R}))$ converges to $f \in H_{n+1}(\mathbb{R})$ if and only if $F(x) = \{A(x) \mid A \in F\}$ converges to $f(x)$ for all $x \in \mathbb{R}$. It is not hard to see that this topology is compatible with the pointwise vector structure.

Definition 2.2. Let $\tilde{f} : \mathbb{R}^n \to \mathbb{R}$ be a function. Define

$$\gamma\left(\tilde{f}\right)(x_1)(x_2)...(x_n) = \tilde{f}(x_1, x_2, ..., x_n).$$

Then $\gamma : \mathrm{Hom}(\mathbb{R}^n, \mathbb{R}) \to H_n(\mathbb{R})$ is clearly a bijection. The function $f = \gamma\big(\tilde{f}\big)$ will be called the *curried version* of $\tilde{f}$.

3. Continuous functions

Proposition 3.1. (see [2], page 34): *Let X, Y be topological spaces such that X is locally compact and Y is regular. On the set of continuous functions $\mathcal{C}(X, Y)$, consider the compact-open topology (see [3]). Then a filter $F \in \mathcal{F}(\mathcal{C}(X, Y))$ converges to a function $f \in \mathcal{C}(X, Y)$ if and only if $F(P) \to f(p)$ whenever $P \to p \in X$. Here, the filter $F(P)$ is based on*

$$\{A(B) \mid A \in F, B \in P\} = \{\{a(b) \mid a \in A, b \in B\} \mid A \in F, B \in P\}.$$

The convergence $F \to f$ so defined is called continuous convergence.

In what follows, we assume that X is a locally compact topological space and Y is a regular topological space.

Proposition 3.2. (see [2], page 27): *If Y is locally compact and Z is another regular topological space, then the map*

$$\gamma : \mathcal{C}(X \times Y, Z) \to \mathcal{C}(X, \mathcal{C}(Y, Z))$$

defined as $\gamma(f)(x)(y) = f(x, y)$, is well-defined and is a homeomorphism.

Proposition 3.3. *Let X be a locally compact topological space and let Y be a Hausdorff topological space (i.e. every filter in Y converges to at most one point). Then $\mathcal{C}(X, Y)$ is also a Hausdorff topological space.*

Proposition 3.4. (see [2], page 29): *Let X be a locally compact space and Y be a regular topological vector space, i.e. a set endowed with both topological and linear structures such that the operations $+ : Y \times Y \to Y$ and $\cdot : \mathbb{R} \times Y \to Y$ are continuous. Then $\mathcal{C}(X, Y)$ is also a topological vector space.*

Proposition 3.5. *Let X, Y be topological spaces such that X is first countable. Then a sequence $\{f_k\}_{k\in\mathbb{N}} \subset \mathcal{C}(X, Y)$ converges to a function $f \in \mathcal{C}(X, Y)$ (in the sense that its derived filter converges) if and only if $f_k(x_k) \xrightarrow[k\to\infty]{} f(x)$ whenever $x_k \xrightarrow[k\to\infty]{} x$ in X.*

> *Proof:* First, let F be the derived filter of $\{f_k\}_{k\in\mathbb{N}}$ and suppose that F converges to a function $f \in \mathcal{C}(X, Y)$. Consider a sequence $\{x_k\}_{k\in\mathbb{N}}$ converging to $x \in X$, with its derived filter P. Now let V be a neighborhood of $f(x)$ in Y. By the definition of continuous convergence, we have $F(P) \to f(x)$, and so $V \in F(P)$. This means that V contains an element of the base of $F(P)$, namely a set
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> $$B_{n_0,m_0} = \{f_n(x_m) \mid n \geq n_0,\ m \geq m_0\}.$$
>
> Hence, for $k \geq \max(n_0, m_0)$, we have $f_k(x_k) \in V$, as desired.
>
> Conversely, suppose that $f_k(x_k) \xrightarrow[k\to\infty]{} f(x)$ whenever $x_k \xrightarrow[k\to\infty]{} x \in X$. First let us show that for any subsequence $\left\{f_{n_k}\right\}_{k\in\mathbb{N}}$, we have $f_{n_k}(x_k) \xrightarrow[k\to\infty]{} f(x)$.
> To this end, let $n \in \mathbb{N}$. We let $z_n = x_{\varphi(n)}$, where $\varphi(n)$ is the minimal number of those $k \in \mathbb{N}$ that satisfy $n \leq n_k$. We clearly see that $n_1 \geq n_2$ implies $\varphi(n_1) \geq \varphi(n_2)$ and that $\varphi(n_k) = k$ for all $k \in \mathbb{N}$. Therefore, the sequence $\{z_n\}_{n\in\mathbb{N}}$ converges to x. Hence we have $f_k(z_k) \xrightarrow[k\to\infty]{} f(x)$, and so

$$f_{n_k}(x_k) = f_{n_k}\left(z_{n_k}\right) \xrightarrow[k\to\infty]{} f(x). \tag{1}$$

Now, to show that $f_k \to f$ in $\mathcal{C}(X, Y)$, we need to show that the derived filter F generated by the sets $C_n = \{f_k \mid k \geq n\}$ converges to f. To this end, let P be a filter in X converging to a point $x \in X$. Note that since X is first countable, the filter P has a countable base $\{A_m\}_{m\in\mathbb{N}}$ of neighborhoods of x. To show that $F(P)$ converges to $f(x)$, assume the contrary. Then some neighborhood V of $f(x)$ does not lie in $F(P)$, which is to say that for all $n, m \in \mathbb{N}$, we have $C_n(A_m) \not\subset V$. That means that for all $n, m \in \mathbb{N}$, there are $k_m \geq m$ and $x_m \in A_m$ such that $f_{k_n}(x_m) \notin V$.
In particular, we have $f_{k_n}(x_n) \notin V$ for all $n \in \mathbb{N}$. At the same time, since the sets A_n generate P, we see that $x_n \xrightarrow[n\to\infty]{} x$, so we must have $f_{k_n}(x_n) \xrightarrow[n\to\infty]{} f(x)$ by (1), a contradiction. ∎

Definition 3.1. Let $n \in \mathbb{N}_0 = \{0, 1, 2, ...\}$. The *space of curried continuous functions* $C_n(\mathbb{R})$ is a Hausdorff topological vector space defined recursively as follows:

- If $n = 0$, we set $C_0(\mathbb{R}) = \mathbb{R}$.
- If $C_n(\mathbb{R})$ is defined, we set $C_{n+1}(\mathbb{R}) = \mathcal{C}(\mathbb{R}, C_n(\mathbb{R}))$. Proposition 3.1, Proposition 3.3, and Proposition 3.4 ensure that $C_{n+1}(\mathbb{R})$ is a Hausdorff topological vector space, provided that $C_n(\mathbb{R})$ is.

Proposition 3.6. *Let* $n \in \mathbb{N}_0$. *Then the map*

$$\gamma : \mathcal{C}(\mathbb{R}^n, \mathbb{R}) \to C_n(\mathbb{R})$$

defined by $\gamma(f)(t_1)(t_2)...(t_n) = f(t_1, ..., t_n)$*, is well-defined and is a homeomorphism.*

Proof: If $n = 0$, the statement is trivial. Assume it is proven for $C_n(\mathbb{R})$. By Proposition 3.2, we have

$$\mathcal{C}(\mathbb{R}^{n+1}, \mathbb{R}) = \mathcal{C}(\mathbb{R} \times \mathbb{R}^n, \mathbb{R}) \cong \mathcal{C}(\mathbb{R}, \mathcal{C}(\mathbb{R}^n, \mathbb{R})) \cong \mathcal{C}(\mathbb{R}, C_n(\mathbb{R})) = C_{n+1}(\mathbb{R}),$$

q.e.d. ∎

4. Differentiable functions

Definition 4.1. Let $n \in \mathbb{N}_0$. The *space of curried differentiable functions* $D_n(\mathbb{R})$ is a Hausdorff topological vector space defined recursively as follows:

- If $n = 0$, we set $D_0(\mathbb{R}) = C_0(\mathbb{R}) = \mathbb{R}$.
- If the space $D_n(\mathbb{R})$ is defined, we define $D_{n+1}(\mathbb{R})$ as the subspace of $\mathcal{C}(\mathbb{R}, D_n(\mathbb{R}))$ consisting of such functions $f : \mathbb{R} \to D_n(\mathbb{R})$, that there is $f' \in C_{n+1}(\mathbb{R})$ which satisfies

$$f(x_0 + h) = f(x_0) + h \cdot f'(x_0) + o(h) \in C_n(\mathbb{R})$$

  for all $x_0 \in \mathbb{R}$.

Proposition 4.1. *Let* $n \in \mathbb{N}_0$. *Then the map*

$$\gamma : C^1(\mathbb{R}^n) \to D_n(\mathbb{R})$$

is well-defined and a homeomorphism. In particular, a function $f \in C_n(\mathbb{R})$ *lies in* $D_n(\mathbb{R})$ *if and only if its counterpart* $\tilde{f} = \gamma^{-1}(f) : \mathbb{R}^n \to \mathbb{R}$ *has continuous partial derivatives.*

Proof: First let us show that γ is well-defined and bijective. ∎

Definition 4.2. Fix $n \in \mathbb{N}_0$. For all $m \in \mathbb{N}$, the *space of curried m times differentiable functions* $D_n^m(\mathbb{R})$ is defined recursively as follows:

- If $m = 1$, we set $D_n^1(\mathbb{R}) = D_n(\mathbb{R})$.
- If $D_n^m(\mathbb{R})$ is defined, we set $D_n^{m+1}(\mathbb{R})$ to be the space of all functions $f \in D_n(\mathbb{R})$ such that $f' \in D_n^m(\mathbb{R})$.

Trivially, for all $n \in \mathbb{N}_0$, we have

$$D_n^1(\mathbb{R}) \supset D_n^2(\mathbb{R}) \supset D_n^3(\mathbb{R}) \supset \ldots$$

Proposition 4.2. *Let $n, m \in \mathbb{N}$. Then a function $f \in C_n(\mathbb{R})$ lies in the differentiability class $D_n^m(\mathbb{R})$ if its counterpart $\tilde{f} = \gamma^{-1}(f) : \mathbb{R}^n \to \mathbb{R}$ lies in the class $C^m(\mathbb{R})$.*

Proposition 4.3. *Let $n, m \in \mathbb{N}$, and $f \in D_n^m(\mathbb{R})$. Then for all $x \in \mathbb{R}$, we have $f(x) \in D_{n-1}^m(\mathbb{R})$.*

Proof: … ∎

Bibliography

[1] N. Bourbaki, *General Topology, Part I*. 1966.

[2] R. Beattie and H.-P. Butzmann, *Convergence Structures and Applications to Functional Analysis*. 2002.

[3] R. H. Fox, "On Topologies For Function Spaces," 1945.